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SH17: A dataset for human safety and personal protective equipment detection in manufacturing industry

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ABSTRACT

Workplace accidents continue to pose significant human safety risks, particularly in the construction and manufacturing industries. The necessity for effective Personal Protective Equipment (PPE) compliance has become increasingly paramount. We focus on developing non-invasive techniques based on the Object Detection (OD) and Convolutional Neural Network (CNN). The aim is to detect and verify the proper use of various types of PPE such as helmets, safety glasses, masks, and protective clothing. This study proposes the SH17 Dataset, consisting of 8,099 annotated images containing 75,994 instances of 17 classes collected from diverse industrial environments, to train and validate the OD models. We have trained state-of-the-art OD models for benchmarking, and initial results demonstrate promising accuracy levels with You Only Look Once (YOLO)v9-e model variant exceeding 70.9% in PPE detection. The validation of the model across cross-domain datasets indicates that integrating these technologies can substantially enhance safety management systems. This approach offers a scalable and efficient solution for industries seeking to comply with human safety regulations while safeguarding their workforce. The dataset is available at https://github.com/ahmadmughees/sh17dataset.

1. Introduction

Occupational Safety and Health (OSH) is a vital multidisciplinary domain focused on ensuring safe working environments and mitigating risks to worker health. It encompasses the prevention of work-related injuries, accidents, and illnesses by systematically managing workplace hazards and promoting health-conscious practices. In industrial manufacturing, PPE plays a pivotal role in protecting workers from various occupational hazards. The importance of PPE lies in its ability to significantly reduce the risks of injuries and illnesses resulting from exposure to hazardous conditions such as toxic chemicals, extreme temperatures, and mechanical dangers.

Extensive research has demonstrated that PPE serves as an essential last line of defense in safeguarding workers when other safety measures are inadequate [1,2]. Manufacturing industries have made considerable efforts to ensure workforce safety by mandating the use of PPE in their facilities, aiming to minimize injury-related incidents that can lead to substantial financial losses. Initially, human inspectors were tasked with monitoring PPE compliance at key points throughout the facility. However, this responsibility has increasingly been augmented by Internet of Thing (IoT) systems, which employ sensor-based methodologies to automate the verification of PPE compliance [3,4].

The advent of Machine Learning (ML) and Computer Vision (CV) systems has prompted a shift towards the development of fully non-invasive solutions for PPE compliance monitoring in manufacturing environments [5,6]. These advancements underscore the potential for more efficient and precise safety monitoring systems. In accordance with the Occupational Safety and Health Administration (OSHA) guidelines [7], Isalovic et al. [8] classified PPE into five categories based on the body regions they are designed to protect: (1) Head, (2) Upper Body, (3) Hands, (4) Feet, and (5) Whole Body. Typical examples of PPE include helmets, safety glasses, face shields, earmuffs, safety vests, gloves, and safety shoes, each tailored to mitigate specific hazards.

While proper selection, usage, and maintenance of PPE are critical for ensuring worker safety, recent studies [9–12] have increasingly utilized ML-based Object Detection (OD) models to detect PPE items across various environments, from construction sites to chemical plants. However, these studies focus on detecting specific PPE items, such as helmets or vests, within controlled environments [5]. Moreover, many of these approaches depend on proprietary or small-scale datasets, which restricts their generalizability and limits their broader applicability.

To overcome these limitations, this study expands the application of OD-based methods for PPE compliance by introducing a large-scale, high-quality, open-source dataset. Our dataset consists of 8099

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Table 2
List of all the annotated classes and their counts.

ID	Name	Additional tags	Instances	Description
1	Person	male, female, children	13 802	Uses visible features for classification.
2	Head	–	11 985	Includes any view of the head: front, back, top or else.
3	Face	–	8950	Only classified as visible when the nose is visible.
4	Glasses	on, off, safety, vision	1945	Detection of safety glasses.
5	Face-mask-medical	on, off	669	Detection of medical face masks
6	Face-guard	on, off	134	Detection of whether faceguards
7	Ear		7730	Focused on the ears for safety equipment detection.
8	Earmuffs	on, off	318	Detection of earmuffs. Over-Ear-Headphones are labeled as earmufs.
9	Hands		15 850	Focus on the hands for safety equipment detection.
10	Gloves	on, off	2790	Detection of gloves.
11	Foot		796	Visible when there are no shoes, and each foot is annotated regardless of with or without socks.
12	Shoes	on, off, safety, other	4560	Shoes detection. Safety: Includes safety shoes and thick joggers. Others: slippers, sneakers, or other types of footwear.
13	Safety-vest	on, off	530	Detection of safety vests.
14	Tools	on, off	4647	Detection of tools being held; on means in hand. If the tool is present in the scene but not in hand, it's off. Pencils and laptops are not considered tools.
15	Helmet	on, off, white, red, black, yellow, blue	927	tags contain the color of a helmet as well.
16	Medical-suit	on, off	155	Detection of medical suits
17	Safety-suit	on, off	530	Detection of safety suits

Note: Tag off means the item is present in the scene but not worn by the person, while on means the item is present and worn by the person.

are illustrated in Fig. 1. The dataset includes the highest number of instances for hands, totaling 15,850, which constitutes 20.9% of the overall instances. The helmet class contains 927 instances, approximately 1.2% of the dataset, while the faceguard class comprises 134 instances, representing the lowest occurrence at 0.2%. A complete list of instances for each class is provided in Table 2. Additionally, Fig. 2 presents the distribution of all classes, their respective instances, and their percentage in the dataset, illustrating an imbalance in class distribution.

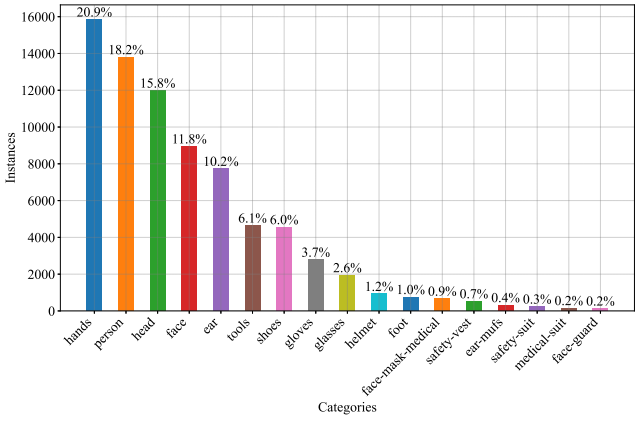

Fig. 2. Distribution of all class instances. The percentage of each class is shown at the top of each bar.

3.5. Evaluation metrics

Object detection models are evaluated using several metrics to assess their performance. Common metrics used by Microsoft Common Objects in Context (MS-COCO) dataset [56] and Pascal VOC dataset [55] include Precision (P), Recall (R), and Mean Average Precision (mAP).

Precision (P) measures the proportion of correctly identified objects among all objects detected by the model and is calculated as

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Recall (R), on the other hand, quantifies the ability of the model to detect all relevant objects in the dataset and is computed as

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Additionally, Mean Average Precision (mAP) is utilized to evaluate the detection accuracy across all classes. This metric is computed based on P and R for each class and subsequently averaged to produce an overall score. The Intersection over Union (IoU) metric is employed to assess the accuracy of object localization, calculating the overlap between the ground truth bounding boxes (b_g) and the model's predicted bounding boxes (b_{pred}) as follows:

$$\text{IoU} = \frac{\text{Area}(b_{pred} \cap b_g)}{\text{Area}(b_{pred} \cup b_g)}$$

where b_g represents the ground truth bounding box, and b_{pred} denotes the bounding box predicted by the OD model. The IoU threshold acts as a filter to remove FP bounding boxes with an IoU score below a certain threshold. This threshold determines the required accuracy for an object to be classified as detected or missed (e.g., IoU ≥ threshold). Different models may use various threshold values, such as 0.25, 0.5, or 0.75, during evaluation.

4. Experimentation and discussion

Multiple OD models were trained using various size variants of YOLOv8, YOLOv9, and YOLOv10 for benchmarking purposes. The dataset was divided into 80% for training and 20% for testing. Default

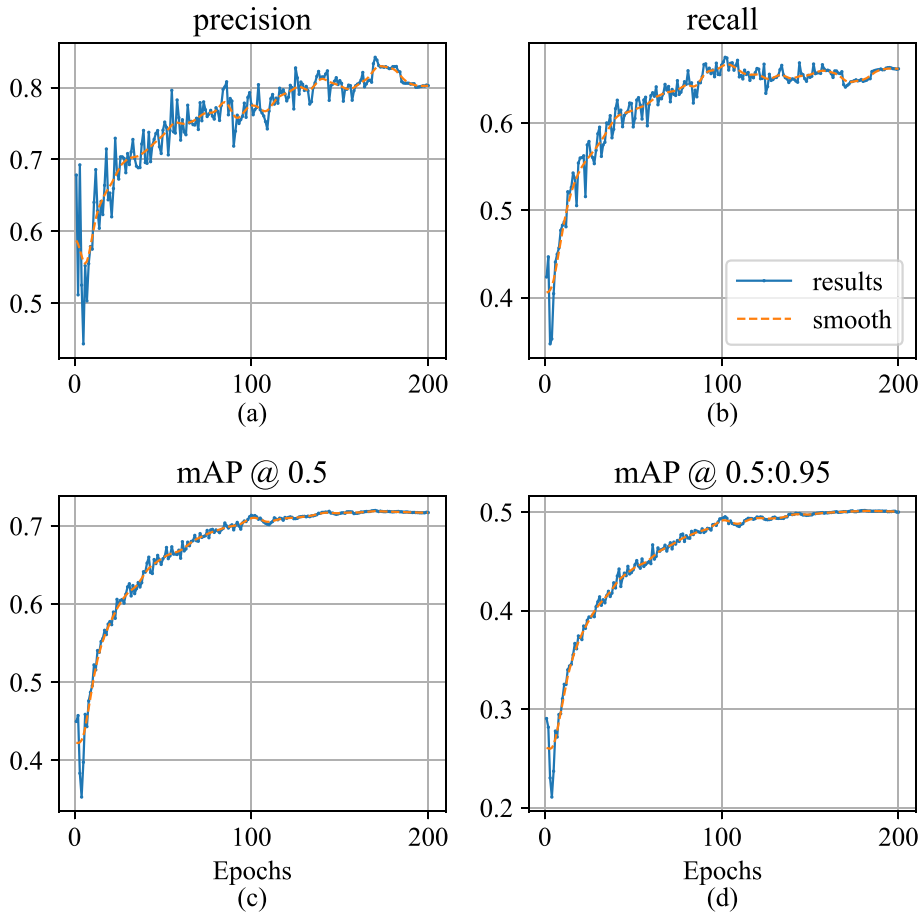

Fig. 3. Training metrics of YOLOv9-e model that performs best among all.

hyperparameters, as suggested by the respective authors in their open-source codebase,[13] were utilized. Transfer learning was employed to adapt a pre-trained model, initially trained on the MS-COCO dataset, which was then trained on the SH17 dataset for 200 epochs. The training hyperparameters were aligned with those presented in [45]. All models were trained on a fixed image size of 640 rather than the original high-quality images due to memory limitations imposed by two NVIDIA RTX GPUs, with batch sizes ranging from 128 for the nano models to 32 for larger scale models. A mosaic of four images, along with horizontal flipping, was employed for data augmentation, and Non-Max-Suppression (NMS) was applied for post-processing the outputs of the YOLOv8 and YOLOv9 model variants. The results of the model training were consolidated in Table 3, evaluated on a separate test set. As noted, YOLOv9-e demonstrated the best performance among all variants, achieving 70.9% mAP@50 and 48.7% mAP50-95. This model comprises 58.1M parameters, in contrast to the next best YOLOv8-x, which contains 68.2M parameters, representing an effective reduction of 15% in the number of parameters. Furthermore, YOLOv9-c exhibited comparable performance to YOLOv8-l while utilizing 32% fewer trainable parameters. Similarly, YOLOv10-x demonstrated comparable performance to YOLOv9-c, with a reduction of 15% in trainable parameters, significantly decreasing both training and inference times.

The training metrics for the YOLOv9-e model, identified as the best-performing model on the SH17 dataset, are presented in Fig. 3. It was observed that the training of the model plateaus at approximately 170 epochs, as the mAP50 does not show any improvement beyond this point. The class-wise accuracy of the YOLOv9-e model on the randomly sampled test set is detailed in Table 4. The hands class, which contains the most samples, demonstrates strong performance with an mAP of 89.8%. The samples for the tools and foot classes exhibit considerable variation in types, including safety shoes, slippers, joggers, and runners, while tools can encompass a wide range of items that a person may be utilizing. This variability has resulted in these classes exhibiting lower mAP scores.

These benchmarks further underscore the results reported in [28, 29,45], where YOLOv9 models demonstrate comparable performance with reduced parameters attributed to the implementation of PGI and GELAN. The YOLOv10 model encounters difficulties with small objects,

Table 3
Comparison of the models trained on SH17. Bold represents the best.

Model	Params	Images	Instances	*P*	*R*	mAP 50	mAP 50-95
Yolo-8-n	3.2	1620	15358	67.5	53.6	58.0	36.6
Yolo-8-s	11.2	1620	15358	81.5	55.7	63.7	41.7
Yolo-8-m	25.9	1620	15358	77.1	60.5	66.6	45.7
Yolo-8-l	43.7	1620	15358	76.7	62.9	68.0	47.0
Yolo-8-x	68.2	1620	15358	77.1	63.1	69.3	47.2
Yolo-9-t	2.0	1620	15358	75.0	52.6	58.5	37.5
Yolo-9-s	7.2	1620	15358	73.6	60.2	65.3	42.9
Yolo-9-m	20.1	1620	15358	77.4	62.0	68.6	46.5
Yolo-9-c	25.5	1620	15358	79.6	60.8	67.7	46.5
Yolo-9-e	58.1	1620	15358	**81.0**	**65.0**	**70.9**	**48.7**
Yolo-10-n	2.3	1620	15358	66.8	53.2	57.2	35.9
Yolo-10-s	7.2	1620	15358	75.8	57.0	62.7	40.9
Yolo-10-m	15.4	1620	15358	71.4	61.4	65.7	43.8
Yolo-10-b	19.1	1620	15358	77.7	59.1	65.8	45.1
Yolo-10-l	24.4	1620	15358	76.0	61.8	67.4	46.0
Yolo-10-x	29.5	1620	15358	76.8	62.8	67.8	46.7

[13] https://github.com/ultralytics/ultralytics

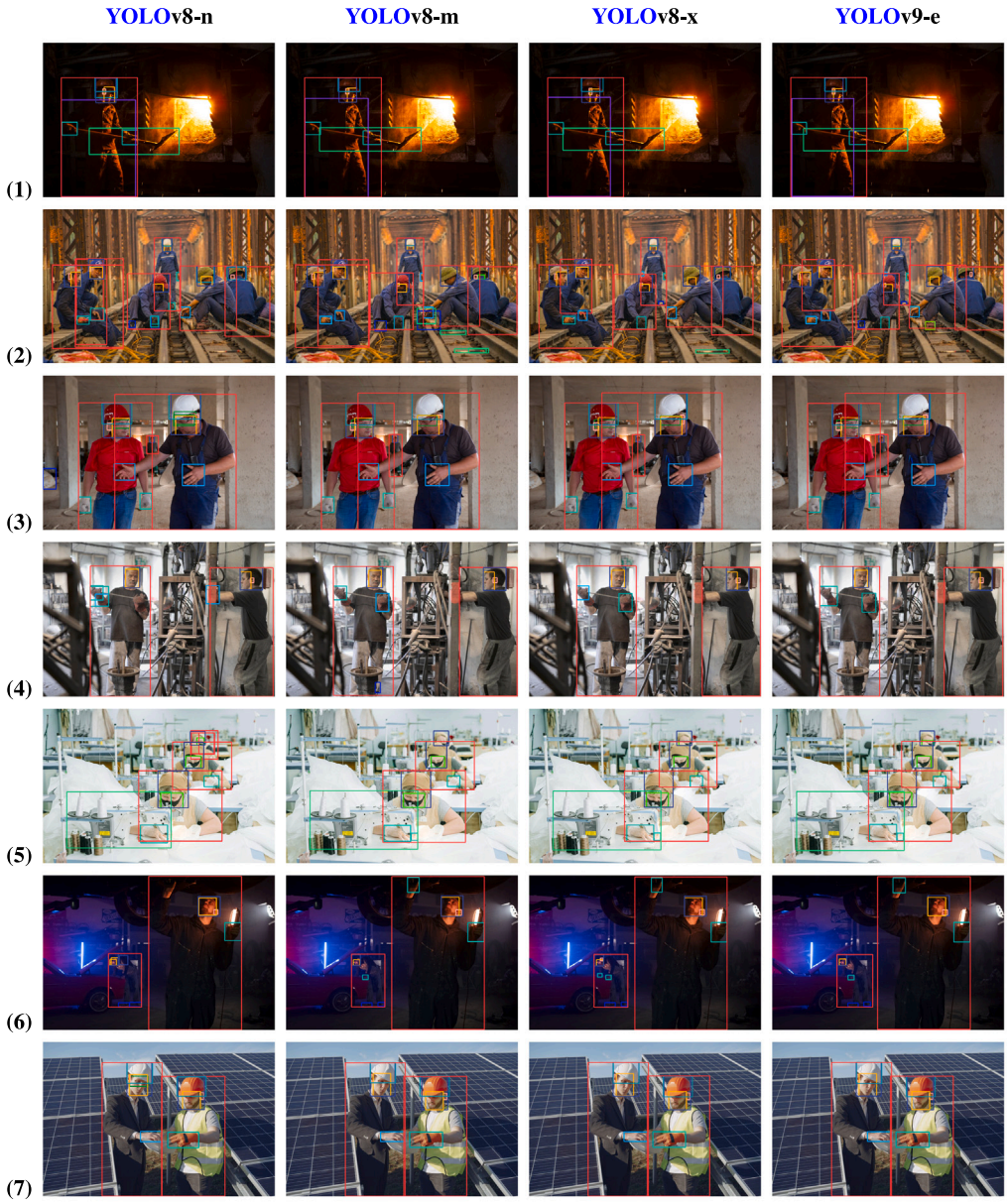

Fig. 4. Visual representation of the predictions made by different models; YOLOv8-n, YOLOv8-m, YOLOv8-x, YOLOv9-e. Best viewed online.

as the SH17 dataset comprises 52% of objects that occupy less than 1% of the area (as explained in Section 3.4). Fig. 4 illustrates some visual differences among the models: YOLOv8-n, YOLOv8-m, YOLOv8-x, and YOLOv9-e. In sample (2) of Fig. 4, both YOLOv8-m and YOLOv8-x made false positive predictions regarding the tool on the ground. For sample (3), the YOLOv8-n model produced a false positive prediction for the face class. In row 5, all models failed to detect the tool held by the worker. The original labeled samples are referenced in Fig. 1.

The efficacy of the model was verified using the SH17 dataset in conjunction with cross-domain datasets. The Pictor-PPE dataset [5] was selected to validate the performance of the YOLOv9-e model. This dataset comprises only three classes: worker, hat, and vest, which were mapped to the Person, Helmet, and Safety-vest classes of SH17, respectively. All publicly available images in the dataset were utilized for validation, resulting in a mAP of 58.8% for YOLOv9-e. This indicates that the trained models can be effectively employed for PPE compliance in manufacturing environments. Although the SH17 dataset contains the fewest samples for the safety-vest class, it still achieves a mAP of 35.5% at 50. The complete results are summarized in Table 5. Fig. 5 presents the confusion matrix for the entire Pictor-PPE dataset as evaluated by YOLOv9-e. It is evident that the person class is occasionally classified as background, while the helmet class is frequently misidentified as the head class. Visual results are also provided in Fig. 6.

5. Conclusion

Ensuring human safety in manufacturing environments is a paramount concern, particularly in maintaining PPE compliance. In this study, the SH17 dataset was introduced and open-sourced, comprising 8099 annotated images across 17 distinct classes that represent various PPE items and body parts. This dataset is freely available for research purposes and aims to advance the field of PPE detection in industrial settings.

To validate the utility of the SH17 dataset, several models were benchmarked, with YOLOv9-e achieving the highest performance. Additionally, the generalization capability of the trained model was tested using the Pictor-PPE dataset, with results indicating that the model is

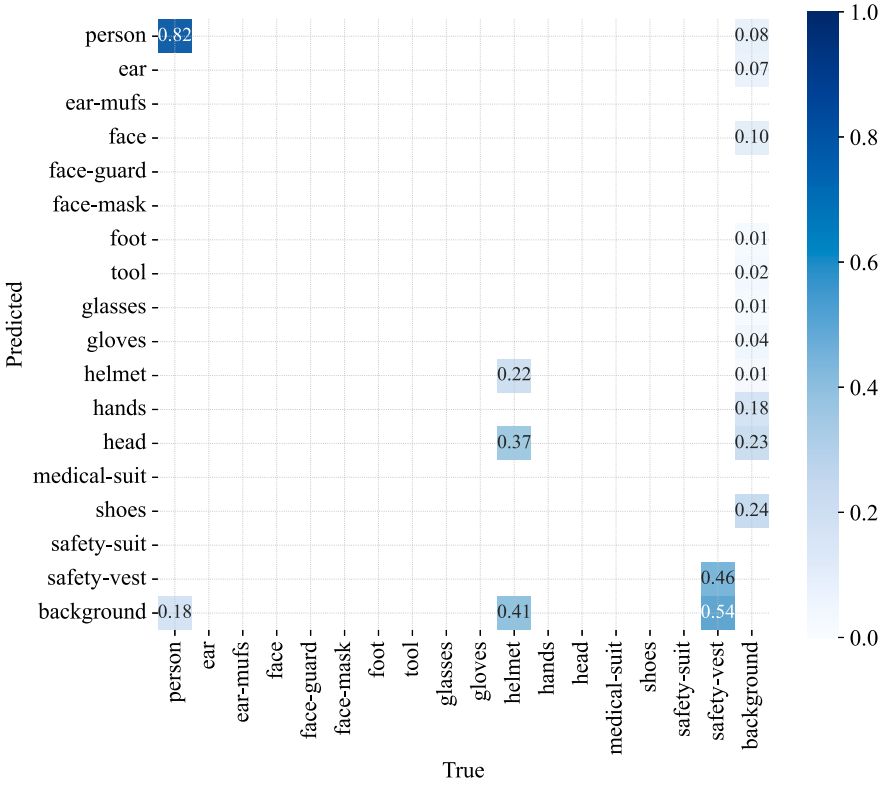

Fig. 5. Confusion Matrix of all the instances of Pictor-PEE dataset.

Fig. 6. Visual results of inference on Pictor-PPE dataset. Best viewed online.

well-suited for deployment in industrial environments to detect and ensure PPE compliance.

Looking ahead, future work could focus on enhancing the detection performance of minority classes within the dataset. This could be accomplished through the development of specialized models and the application of customized training methodologies. Such advancements are anticipated to further enhance the applicability of PPE detection systems in diverse and challenging industrial environments.

Table 4
Class-wise accuracy of YOLOv9-e model.

Class	Images	Instances	P	R	mAP 50	mAP 50-95
all	1620	15 358	81.0	65.0	70.9	48.7
hands	1284	3212	91.4	83.9	89.8	64.8
person	1515	2734	90.9	89.2	92.1	77.9
head	1314	2427	94.8	89.1	93.5	74.3
face	1155	1855	96.0	88.1	93.8	73.8
ear	987	1612	91.2	75.5	84.3	55.0
shoes	320	956	79.2	62.8	70.8	43.2
tool	455	923	67.4	39.1	43.2	27.6
gloves	254	529	81.6	58.9	66.5	43.5
glasses	323	398	87.4	72.6	76.4	46.9
helmet	93	154	81.3	67.8	77.0	57.6
face-mask	75	151	88.8	73.2	75.5	49.2
foot	64	149	51.7	22.1	29.3	14.0
safety-vest	45	97	66.4	55.0	57.7	38.1
ear-mufs	38	49	79.6	46.9	57.1	40.5
safety-suit	28	45	65.7	53.3	58.5	38.3
medical-suit	30	43	86.0	65.1	68.5	40.6
face-guard	23	24	76.8	62.5	71.7	42.8

Table 5
Class-wise accuracy of YOLOv9e model on the Pictor-PPE dataset [5].

Class	Images	Instances	P	R	mAP 50	mAP 50-95
all	654	3,477	73.5	50.3	58.9	37.6
person	654	2,080	83.6	80.7	85.5	61.6
helmet	451	1,369	94.6	23.7	55.5	30.0
safety-vest	12	28	42.4	46.4	35.5	21.2

Acronyms

AI	Artificial Intelligence.
CIoU	Complete Intersection Over Union.
CNN	Convolutional Neural Network.
COVID-19	Coronavirus Disease.
CV	Computer Vision.
DFL	Distribution Focal Loss.
FP	False Positive.
GELAN	Generalized Efficient Long-Range Attention Network.
IoT	Internet of Thing.
IoU	Intersection over Union.
mAP	Mean Average Precision.
ML	Machine Learning.
MS-COCO	Microsoft Common Objects in Context.
NMS	Non-Max-Suppression.
NSERC	Natural Sciences and Engineering Research Council of Canada.
OCI	Ontario Center of Innovation.
OD	Object Detection.
OSH	Occupational Safety and Health.
OSHA	Occupational Safety and Health Administration.
P	Precision.
PGI	Programmable Gradient Information.
PPE	Personal Protective Equipment.
R	Recall.
R-CNN	Region Proposal Convolutional Neural Network.
RoI	Region of Interest.
SSD	Single Stage Detector.
TP	True Positive.
YOLO	You Only Look Once.

Table A.1
Metadata format for the dataset.

Field	Description
Unique Identifier	Unique code for each image for identification.
Width and Height	Dimensions of the image, specified in pixels.
URL	Web address where the image can be accessed.
Photographer Name	Name of the individual who captured the image.
Photographer URL	Web address to the photographer's portfolio or profile.
Photographer ID	Unique identifier for the photographer, used for reference or database purposes.
Average Color	Average color in the hexadecimal code.
Source	Origin from which the image was obtained.
Liked	Indicates whether the image has been marked as liked or favored on the online platform, a boolean value.
Description	Summary of the image's content, context, or theme.

CRediT authorship contribution statement

Hafiz Mughees Ahmad: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Afshin Rahimi:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis.

Institutional review

Not applicable.

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Ethics statements

Not applicable because this work does not involve the use of animal or human subjects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. SH17 dataset

A.1. Meta data

The collected meta-data for each image has also been provided, which can be utilized to reconstruct the dataset from the source or can serve as supplementary data during model training. The list of objects and their definitions included in the meta-data is detailed in Table A.1. Additionally, scripts for constructing the data from the source are available in the SH17 dataset GitHub repository.

Table A.2
Quantitative analysis of the additional tags with Persons class.

	Male	Female	Children	Total
White	2,432	2,032	37	4,501
Black	1,098	776	7	1,881
Brown	577	218	12	807
Asian	963	1,272	52	2,287
Total	5,070	4,298	1,08	9,476

Table A.3
Sorted list of instances in each class in Train and Test Set.

Class	Train	Test
face-guard	110	24
medical-suit	114	43
safety-suit	195	45
ear-mufs	269	49
safety-vest	433	97
face-mask-medical	519	151
foot	610	149
helmet	773	154
glasses	1,547	398
gloves	2,261	529
shoes	3,604	956
tools	3,724	923
ear	6,118	1,612
face	7,095	1,855
head	9,558	2,427
person	11,068	2,734
hands	12,638	3,212

A.2. Person demographics

In addition to the bounding boxes representing the Persons class, additional tags based on visible features within the images have been included. While these tags are primarily derived from observable characteristics, some inaccuracies may be present, as visual features are not consistently reliable indicators. Nonetheless, these tags have been incorporated to identify and mitigate potential biases related to gender or ethnic representation. The tags encompass categories such as male, female, children, and various ethnic backgrounds, including Black, Brown, White, and Asian. A quantitative analysis of these tags is provided in Table A.2.

A.3. Train test split

In this study 80% of the data was allocated for training purposes, while 20% was designated for testing. A separate list of the training and test files is provided in the SH17 dataset GitHub repository. The complete list of instances in both the training and test sets is presented in Table A.3.

Appendix B. Supplementary materials

Supporting data is available at SH17 dataset GitHub repository.

Data availability

The data presented in this study are available in SH17 dataset GitHub repository at https://github.com/ahmadmughees/sh17dataset.

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